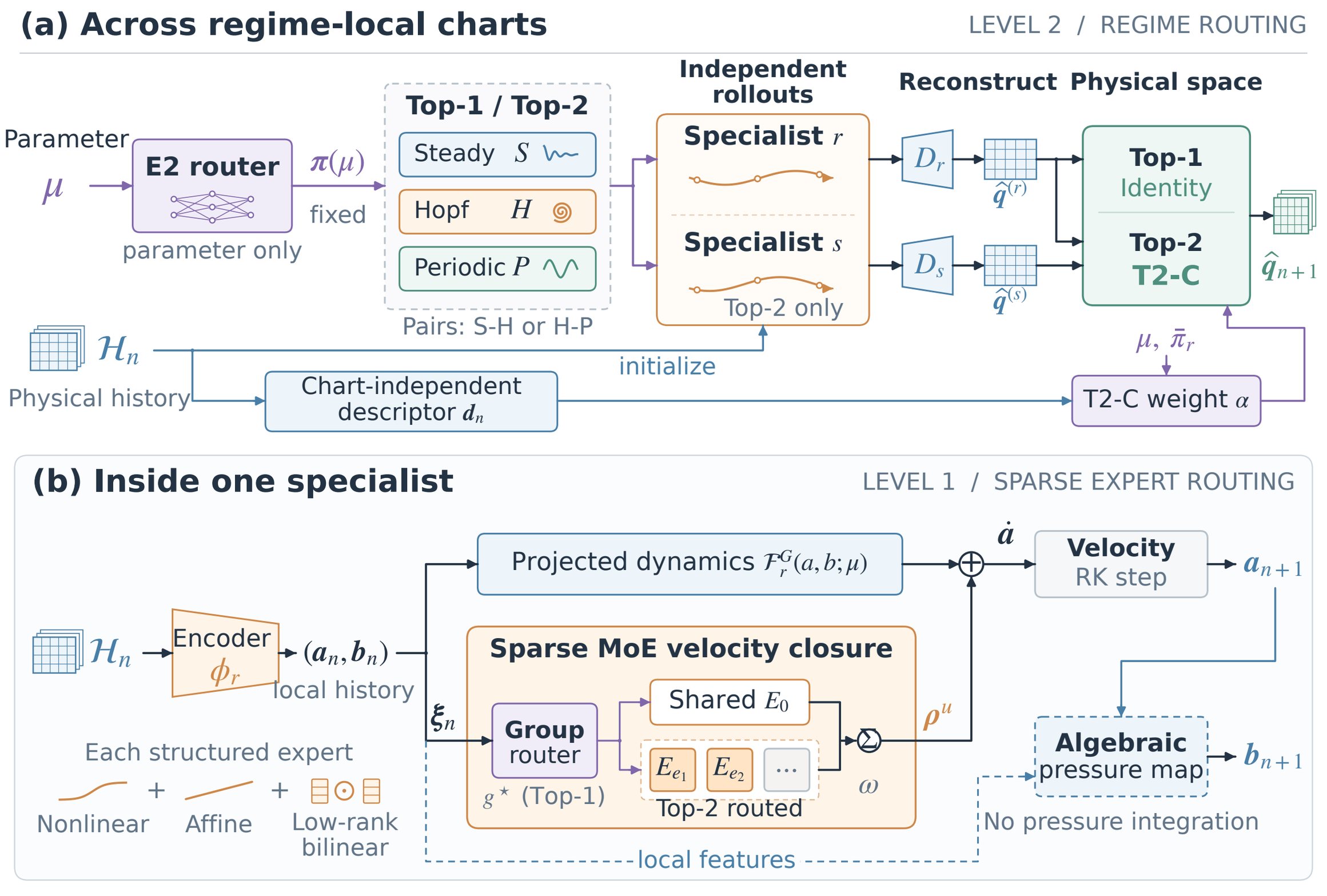

(a) Across regime-local charts
LEVEL 2 / REGIME ROUTING
Parameter
μ
E2 router
parameter only
$\pi(\mu)$
fixed
Top-1 / Top-2
Steady S
Hopf H
Periodic P
Pairs: S-H or H-P
Independent rollouts
Specialist r
Specialist s
Top-2 only
Reconstruct Physical space
D_r
D_s
$\hat{q}^{(r)}$
$\hat{q}^{(s)}$
Top-1
Identity
Top-2
T2-C
$\hat{q}_{n+1}$
$\mathcal{H}_n$
Physical history
initialize
Chart-independent descriptor d_n
$\mu, \bar{\pi}_r$
T2-C weight α
(b) Inside one specialist
LEVEL 1 / SPARSE EXPERT ROUTING
Projected dynamics $\mathcal{F}_r^G(a,b;\mu)$
$\dot{a}$
Velocity
RK step
a_{n+1}
$\mathcal{H}_n$
Encoder ϕ_r
(a_n, b_n)
local history
Sparse MoE velocity closure
ξ_n
Group router
$g^\star$ (Top-1)
Shared E_0
E_{e_1}
E_{e_2}
...
Top-2 routed
ω
ρ^u
Each structured expert
Nonlinear
+
Affine
+
Low-rank bilinear
Algebraic pressure map
b_{n+1}
No pressure integration
local features